Suggestions to Strengthen the Digital Tax and Product Tracking System

Rwanda Revenue Authority · Integrated EBM Complement System

Based on the full system design covering EBM gaps, price underreporting prevention, AI price suggestion, and biometric customer confirmation, the following ten suggestions address the core remaining vulnerabilities. Each links a specific problem in the system to a concrete, implementable solution.

THE TEN SUGGESTIONS

1 Merchants declare lower prices than they charge

Suggested solution

Layer three defences simultaneously

Combine the AI price floor (blocks suspicious entries before submission), biometric customer confirmation (the customer signs the actual price at the moment of sale), and mobile money cross-referencing (actual payment flows cannot lie). Any one of these alone can be worked around. All three together make price fraud structurally very difficult — each layer compensates for the weaknesses of the others.

2 Traders operating without TIN numbers

Suggested solution

Bring TIN registration to the merchant's device

Link TIN registration directly to Rwanda's Irembo platform so a trader can register and receive a TIN on the same device they use to sell, in under 10 minutes, without visiting a physical office. Offer a 30-day grace period for existing informal traders to register before enforcement begins. Remove friction — when registration is easy, the excuse of not having a TIN disappears.

3 Undeclared stock that never enters the system

Suggested solution

Lock goods into the system at their origin point

Every factory and importer must declare production or import quantities before goods leave their premises. Connect directly to Rwanda's customs database so imported goods are activated in the system at the border with quantities locked. A product that has no system record cannot be legally restocked by any registered merchant, because restocking requires a valid source transaction ID. Attack the problem at entry, not downstream.

4 Limited visibility of goods across the supply chain

Suggested solution

Introduce a product passport for every batch

Every batch of goods receives a unique lot code that travels through the chain. When a wholesaler buys from a factory, that lot transfers. When a retailer buys from the wholesaler, it transfers again. RRA can at any moment ask: 'Where is lot 0047 of Kigali Cement's March production?' and trace it through every hand it has passed through. Ghost stock — declared in one place but sold elsewhere — becomes immediately detectable.

5 Biometric confirmation is impractical for every small transaction

Suggested solution

Apply biometrics selectively based on transaction value

For transactions below a threshold such as 5,000 RWF, skip biometric confirmation and use a simplified SMS receipt instead. Focus biometric confirmation on mid-to-large transactions where the tax evasion risk is meaningful. This makes the system practical without abandoning the principle — the friction is applied where the risk is highest, not uniformly across all transactions regardless of size.

6 Merchants collude to set artificially low declared prices

Suggested solution

Anchor the AI benchmark to external, uncorruptible price sources

If local merchants coordinate to declare false prices, the AI peer benchmark learns from manipulated data. The solution is to anchor the reference not just to local merchant declarations, but to independent external sources: national statistics office data, supermarket price feeds from Nakumatt and Simba, and import/customs declared values. These external sources cannot be manipulated by local merchants and give the AI a ground truth resistant to collusion.

7 Honest merchants suffer the same friction as fraudulent ones

Suggested solution

Create a compliance passport with real commercial privileges

Merchants who maintain a green compliance score for six or more months receive streamlined transactions: lighter confirmation steps, faster restock approvals, and reduced audit frequency. Fraud risk is concentrated in non-compliant merchants, so direct the friction there. A compliant merchant should feel rewarded, not burdened. This also creates a commercial incentive for compliance beyond just avoiding penalties.

8 No historical data to learn from at system launch

Suggested solution

Seed the AI with external reference prices from day one

When the system first launches, there is no transaction history and no peer benchmark. Seed the AI using Ministry of Trade price surveys, import manifests, and supermarket price APIs so it has a credible baseline from day one. Gradually replace these external seeds with real transaction data as it accumulates over the first three to six months. Never start the system with an empty reference — that creates a window where any declared price is accepted.

9 Privacy concerns and public distrust of biometric collection

Suggested solution

Design trust as a core system requirement, not an afterthought

Run a clear public communication campaign before launch, in Kinyarwanda, explaining exactly what is stored (a mathematical hash, not the fingerprint image itself), who can access it (only RRA with a court order), and what the customer gains (a guaranteed receipt, protection against being overcharged, mobile money rewards for fraud reporting). Without public trust, customers will refuse to confirm transactions and the biometric layer collapses. Trust must be designed in from the beginning.

10 Unreliable internet connectivity in rural markets

Suggested solution

Make offline-first the default architecture, not an exception

Design every merchant device to work at zero connectivity and treat online as an enhancement, not a requirement. Transactions are stored encrypted locally and synced when connectivity is restored. The biometric matching uses a locally cached template for Tier 1 transactions, with a reconciliation check on reconnection. Unsynced transactions older than 48 hours are automatically flagged for review. The system must serve rural Rwanda as reliably as it serves Kigali.

IMPLEMENTATION PRIORITY SUMMARY

The ten suggestions fall into three implementation phases. Start with the highest-impact items that address the most fundamental gaps before adding the more sophisticated layers.

Phase	Suggestion	Why first
Phase 1 Immediate	Mobile money integration (#1) External price seeding (#8) TIN via Irembo (#2)	Hardest to fake, cheapest to deploy, no hardware required
Phase 2 3-6 months	Customs / origin locking (#3) Product lot passport (#4) Offline-first architecture (#10)	Requires integration with govt. systems and device firmware updates
Phase 3 6-12 months	Compliance passport (#7) External price anchoring (#6) Value-tiered biometrics (#5) Public trust campaign (#9)	Requires sufficient data history and public engagement program

THE SINGLE MOST IMPORTANT SUGGESTION

Integrate with mobile money from the beginning, not as an add-on later

Rwanda has one of the highest mobile money penetration rates in Africa. Most merchants already receive payments through MTN Mobile Money or Airtel Money. If the system can read those incoming payment flows — with proper legal authorization — declared revenue becomes verifiable against real money received. A merchant who declares 300,000 RWF in monthly sales but received 900,000 RWF in mobile payments cannot hide that gap. No biometric, no AI price range, no audit can match the certainty of following the actual money. Everything else in this system makes fraud harder. Mobile money integration makes the fraud visible even after it happens.

These ten suggestions, applied in phases, transform the system from a passive recording tool into an active, self-defending compliance infrastructure — one that makes tax evasion structurally difficult, detectable in real time, and costly enough that honest compliance becomes the easier path for every merchant.